

Paul Zinn-Justin

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CURRICULUM VITAE

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PROFESSIONAL EXPERIENCE

2020–	Professor at the School of Mathematics and Statistics of the University of Melbourne (Australia).
2016–2020	ARC Future Fellow at the School of Mathematics and Statistics of the University of Melbourne (Australia).
2013–2016	Directeur de recherches (senior scientist) CNRS at LPTHE (UMR 7589), Université Pierre et Marie Curie (France).
2012	“Research Member” of the MSRI program “Random Spatial Processes” (Berkeley).
2009–2010	“Visiting Professor” of the KNAW (Royal Netherlands Academy of Arts and Sciences), visiting the University of Amsterdam.
2008–2013	Chargé de recherches CNRS at LPTHE (UMR 7589), Université Pierre et Marie Curie (France).
2005–2008	Chargé de recherches CNRS at LPTMS (UMR 8626), Université Paris-Sud (France).
2004–2005	Chargé de recherches CNRS at Laboratoire Poncelet (LIFR-MIIP, UMI 2615), Moscow (Russia).
2000–2004	Chargé de recherches (junior scientist) CNRS at LPTMS (UMR 8626), Université Paris-Sud (France).
2000	Visitor at C.N. Yang Institute for Theoretical Physics, SUNY Stony Brook.
1999–2000	Post-doctoral fellow at the New High Energy Theory Center, Rutgers University.
1998–1999	National service.
1995–1998	PhD in theoretical physics at the Laboratoire de Physique Théorique de l’Ecole Normale Supérieure de Paris, supervisor V. Kazakov.

RESEARCH INTERESTS

- Algebraic combinatorics, symmetric functions; Schubert calculus, enumerative geometry.
- Exactly solvable models in statistical mechanics, quantum integrability, Yang–Baxter equation, and quantum groups.
- Random matrices, and connections between mathematical physics and enumerative combinatorics.
- Computational and experimental mathematics, including algorithm design, computer algebra and machine learning.

PRIZES, AWARDS

2016–2020	Australian Research Council Future Fellowship “From quantum integrable systems to algebraic geometry and combinatorics”.
2011–2016	European Research Council grant “Loop models, Integrability and Combinatorics”.
2010–2014	Prime d’excellence scientifique (scientific excellency award) from CNRS.
2009	Visiting Professorship award from the Royal Netherlands Academy of Arts and Sciences.

TEACHING EXPERIENCE

2019–2026	<i>Discrete Mathematics, Linear Algebra, Group theory and linear algebra, Enumerative combinatorics, Exactly solvable models, Experimental mathematics</i> (48h/36h) courses at the University of Melbourne.
2022	LAWRGe lectures at the University of Southern California: <i>Schubert Calculus and Quantum Integrability</i> (14h).
2019	Mini-course at the University of Virginia: <i>Quantum integrability and symmetric polynomials</i> (10h).
2019	Mini-course at the University of Melbourne: <i>Schubert calculus</i> (12h).
2018	Mini-course at the University of Melbourne: <i>Geometry, Integrability and symmetric functions</i> (6h).
2015	Lectures at the Department of Mathematics of the Higher School of Economics (Moscow): <i>Geometry, Integrability and symmetric functions</i> (8h).
2013–2015	<i>Intégrabilité quantique</i> , Masters course at Université Pierre et Marie Curie (30h).
2012	Chern–Simons lectures at the University of California at Berkeley: <i>Exactly solvable tiling models, Schur functions and the Littlewood–Richardson rule</i> (6h).
2008	Course at Les Houches summer school: <i>Integrability and combinatorics: selected topics</i> .
2007	Lectures at Barcelona University: <i>Quantum integrability and Razumov–Stroganov type conjectures</i> .
1997–1998	Teaching Assistant at ENS Paris: mathematics course.
1995–1997	Oral examiner at preparatory school Louis-Le-Grand (bachelor level).

STUDIES AND ACADEMIC HONOURS

2008	Habilitation à Diriger des Recherches “Six-Vertex, Loop and Tiling models: Integrability and Combinatorics” (Université Pierre et Marie Curie, Paris 6).
1998	PhD “Quelques applications de l’Ansatz de Bethe”, mention très honorable avec les félicitations du jury (highest mark) (Université Pierre et Marie Curie, Paris 6).
1996	Mathematics “Agrégation” (teaching diploma).
1994–1995	D.E.A. (Master’s) in theoretical physics, ENS.
1993–1994	Licence et maîtrise in mathematics, licence in physics (Bachelor’s degrees).

- 1993 Admission at Ecole Normale Supérieure (ENS, Paris) on the Mathematics–Physics–Computer Science entrance exam.
- 1992 Silver medal at the International Mathematical Olympiads.

ADMINISTRATION OF RESEARCH

- Organisation of scientific meetings:
 - conference “Formal Power Series and Algebraic Combinatorics 2026”, Seattle (Program committee member).
 - conference “At the crossroads of physics and mathematics: the joy of integrable combinatorics — a conference in honour of Philippe Di Francesco’s 60th birthday”, 24–26 June 2024, Saclay (France).
 - conference “Integrability, Combinatorics and Representations ’19”, 2–6 September 2019, Giens (France).
 - workshop “Geometric R -matrices”, 18–22 December 2017, MATRIX (Australia).
 - program “New trends in integrable models”, 15 August–25 November 2016, International Institute of Physics (Natal, Brazil).
 - program “Statistical Mechanics, Integrability and Combinatorics”, 11 May–3 July 2015, and conference “Lattice Models: Exact Methods and Combinatorics”, 18–22 May 2015, Galileo Galilei Institute (Florence, Italy).
 - conference “Integrability and Combinatorics ’14”, Giens (France), 23–27 June 2014.
 - conference “Formal Power Series and Algebraic Combinatorics 2013”, Paris (Program committee member).
 - conference “Random Matrices and Integrable Systems”, Les Houches, 5–9 March 2012.
 - conference “GranMa ’11”, IHP (Paris), 3–6 October 2011.
 - conference “Geometry and Integrability in Mathematical Physics ’08”, CIRM Luminy (France), 15–19 September 2008.
 - conference “Geometry and Integrability in Mathematical Physics ’06”, Moscow (Russia), 15–19 June 2006.
 - workshop “Combinatorial methods in Physics and Knot Theory”, Moscow (Russia), 21–25 February 2005.
- Supervision of six PhD students (T. Fonseca 2007–2010, A. Garbali 2012–2015, A. Gunna 2020–2024, A. Stoyan 2023–2026, Junru Hu 2026–, Jakob Sloan 2026–), of ten post-docs (L. Cantini 2006–2008, K. Kalle 2007–2008, M. Wheeler and D. Betea 2012–2014, A. Ponsaing 2013–2014, P. Finch 2014–2015, I. Halacheva 2018–2020, J. Lamers 2020–2021, J.-E. Bourgine 2022–2023, L. Cassia 2023–2025). L. Cantini later received the CNRS Bronze Medal. Twelve of these are still in academia, among whom five have permanent positions; the remaining four successfully transitioned to the private sector. Supervision of seven Master students and three Bachelor students summer internships.
- Recipient of four Discovery Projects “Multivariate polynomials: combinatorics and applications” (2014–2017), “Matrix product multi-variable polynomials from quantum algebras” (2019–2023), “Elliptic Schubert calculus” (2021–2026), “Shuffle algebras and vertex models” (2024–2026), and a Future Fellowship “From quantum integrable systems to algebraic geometry and combinatorics” (2016–2020) from the ARC.
- Recipient of an ERC consolidator grant for the project “Loop models, Integrability and Combinatorics”, 2011–2016 (budget: 900k€).
- “Scientist in charge” of the Marie Curie Research Training Network “ENRAGE” (Random Geometry and Random Matrices: From Quantum Gravity to Econophysics), 2005–2009.

- Coordinator of the ANR program “GIMP” (Geometry and Integrability in Mathematical Physics), 2006–2009.
- Former editor for *Communications in Mathematical Physics*; editor for *Algebraic Combinatorics*.
- Referee for numerous journals (Commun. Math. Phys., Lett. Math. Phys., Electronic J. of Combinatorics, Adv. Appl. Math., J. of Combin. Theory A, JSTAT, J. of Physics A, etc); expert for several research organisations (ANR – France, NWO – Netherlands, ESF – Europe, ARC – Australia).
- Member of the Committee of Academic Sponsors of SLMATH (Berkeley). Former member of the Diversity and Inclusion committee (Melbourne University). Former member of the laboratory council (LPTMS and LPTHE).
- PhD referee (A. Ponsaing, L. Poulain d’Andecy).
- Member of the European Marie Curie Research Training Network “ENIGMA”, 2005–2008, of the ESF program “MISGAM”, 2005–2009, and of the ANR program “GranMa”, 2009–2011.

SUMMARY OF PUBLICATIONS

- 80 publications in major international peer-reviewed journals.
- 7 chapters of books.
- 8 preprints.
- 1 popular science article.
- “Habilitation à Diriger des Recherches” thesis was published as a monograph.
- h-index: 33 (Google Scholar, April 2026)

SUMMARY OF CONFERENCES

- Organised conferences: 12
- Invited oral presentations in international conferences and schools: 100
- Other oral presentations: 69

SELECTED INVITED AND PLENARY CONFERENCE TALKS

- Conference “FPSAC”, UC Davis (USA), 17–21 July 2023: *Schubert puzzles as exactly solvable models* (plenary talk).
- International Congress of Mathematical Physics, Montreal (Canada), 23–28 July 2018: *From quantum integrability to Schubert calculus*.
- Conference “Integrability and Algebraic Combinatorics”, IPAM UCLA (USA), 15–19 April 2024: *The MacNeille completion of the type B Bruhat poset*.
- Conference “Quantum integrability and quantum Schubert calculus”, Kavli Royal Society Centre (UK), 11–13 June 2018: *Schubert puzzles and quantum integrability*.
- Symposium FoPM Tokyo (Japan), 6–8 February 2023: *From exactly solvable models of statistical mechanics to combinatorics*.
- Main speaker at the 76th “Séminaire Lotharingien de Combinatoire”, 3–6 April 2016: *Razumov–Stroganov Correspondences and the Geometry of Schubert Varieties*.

- Professional expertise in algorithm design.
- High level programming skills in:
 - Traditional programming languages: python, c, c++, C#, Lisp, assembly language. . . ;
 - Web-based languages and related: HTML/CSS/JavaScript/TypeScript, PHP/SQL;
 - Mathematics-related languages: Macaulay2, Mathematica.
- Contributes to various open source projects.
Developer for (since 2025, “author of”) the open source software “Macaulay2”:
 - Bug fixes, implementation of new functions.
 - Rewrite of the output engine.
 - Macaulay2Web, a web-based interface using the $\text{K}\text{A}\text{T}\text{E}\text{X}$ technology.
- Contributes entries to the On-Line Encyclopedia of Integer Sequences.
- Various projects can be found on my website and on my GitHub page.

PUBLICATIONS

- [96] Paul Zinn-Justin. q -difference equations for two classes of pattern avoiding permutations, 2026. [arXiv:2608.09078](#).
- [95] Geordie Williamson, Oded Yacobi, and Paul Zinn-Justin. Generating Hadamard matrices with transformers, 2026. [arXiv:2604.11101](#).
- [94] Ajeeth Gunna, Michael Wheeler, and Paul Zinn-Justin. Inhomogeneous q -Whittaker polynomials I: Duality and expansions, 2025. [arXiv:2512.04468](#).
- [93] Allen Knutson and Paul Zinn-Justin. Hybrid pipe dreams for the lower-upper scheme, 2025. [arXiv:2509.01857](#).
- [92] Ajeeth Gunna, Michael Wheeler, and Paul Zinn-Justin. Structure constants for spin Hall–Littlewood functions, 2025. [arXiv:2504.19205](#).
- [91] Bruce W. Westbury and Paul Zinn-Justin. A uniform R -matrix for adjoint representations. *Bulg. J. Phys.*, 52(1):159–164, 2025.
- [90] Allen Knutson and Paul Zinn-Justin. Generic pipe dreams, lower-upper varieties, and Schwartz–MacPherson classes. In *Proceedings of the 37th Conference on Formal Power Series and Algebraic Combinatorics (San Diego, 2006)*, volume 93B, 2025. [arXiv:2411.11208](#).
- [89] Bruce W. Westbury and Paul Zinn-Justin. A uniform trigonometric R -matrix for the exceptional series, 2024. [arXiv:2406.01348](#).
- [88] Paul Zinn-Justin. Integrability and combinatorics. In *Encyclopedia of Mathematical Physics*, volume 3. 2025. chapter of the Encyclopedia of Mathematical Physics. [arXiv:2404.13221](#).
- [87] Yaping Yang and Paul Zinn-Justin. Higher spin representations of the Yangian of $\mathfrak{sl}_2$ and R -matrices, 2024. [arXiv:2403.17433](#).
- [86] Paul Zinn-Justin. The *CotangentSchubert* Macaulay2 package. *Journal of Software for Algebra and Geometry*, 14:73–85, 2024. <https://www.unimelb-macaulay2.cloud.edu.au/#tutorial-CotangentSchubert>.
- [85] Allen Knutson and Paul Zinn-Justin. Schubert puzzles and integrability III: separated descents, 2023. [arXiv:2306.13855](#).
- [84] Alexandr Garbali and Paul Zinn-Justin. Shuffle algebras, lattice paths and the commuting scheme. In *Hypergeometry, Integrability and Lie Theory*. 2022. Editors Erik Koelink, Stefan Kolb, Nicolai Reshetikhin and Bart Vlaar. [arXiv:2110.07155](#).

- [83] Allen Knutson and Paul Zinn-Justin. Schubert puzzles and integrability II: multiplying motivic Segre classes. *Comm. A.M.S.*, 6:68–152, 2026. [arXiv:2102.00563](#).
- [82] Ajeeth Gunna and Paul Zinn-Justin. Vertex models for canonical Grothendieck polynomials and their duals. *Algebraic Combinatorics*, 6(1):109–163, 2023. [arXiv:2009.13172](#).
- [81] Andrew Elvey Price and Paul Zinn-Justin. The six-vertex model on random planar maps revisited. *Journal of Combinatorial Theory, Series A*, 196:105739, 2023. [arXiv:2007.07928](#).
- [80] Paul Zinn-Justin. The trigonometric E_8 R-matrix. *Lett. Math. Phys.*, 110:3279–3305, 2020. [arXiv:2004.02044](#).
- [79] Paul Zinn-Justin. Honeycombs for Hall polynomials. *Electron. J. Combin.*, 27:P2.23, 2020. [arXiv:1909.10720](#).
- [78] Dmitri Bykov and Paul Zinn-Justin. Higher spin $\mathfrak{sl}_2$ R-matrix from equivariant (co)homology. *Lett. Math. Phys.*, 110:2435–2470, 2020. [arXiv:1904.11107](#).
- [77] Mireille Bousquet-Mélou, Andrew Elvey Price, and Paul Zinn-Justin. Eulerian orientations and the six-vertex model on planar maps. In *Proceedings of the 31st Conference on Formal Power Series and Algebraic Combinatorics (Ljubljana, 2019)*, volume 82B, 2020. <http://fpsac2019.fmf.uni-lj.si/resources/Proceedings/129.pdf>, [arXiv:1902.07369](#).
- [76] Iva Halacheva, Allen Knutson, and Paul Zinn-Justin. Restricting Schubert classes to symplectic Grassmannians using self-dual puzzles. In *Proceedings of the 31st Conference on Formal Power Series and Algebraic Combinatorics (Ljubljana, 2019)*, volume 82B, page Art. 83, 2020. <http://fpsac2019.fmf.uni-lj.si/resources/Proceedings/154.pdf>, [arXiv:1811.07581](#).
- [75] Andrew Conway, Anthony Guttman, and Paul Zinn-Justin. 1324-avoiding permutations revisited. *Advances in Applied Mathematics*, 96:312–333, 2018. [arXiv:1709.01248](#).
- [74] Allen Knutson and Paul Zinn-Justin. Schubert puzzles and integrability I: invariant trilinear forms. *Comm. A.M.S.*, 6:1–67, 2026. [arXiv:1706.10019](#).
- [73] Paul Zinn-Justin. Loop models and K-theory. *SIGMA Symmetry Integrability Geom. Methods Appl.*, 14:paper 69, 48, 2018. [arXiv:1612.05361](#).
- [72] Allen Knutson and Paul Zinn-Justin. Grassmann–Grassmann conormal varieties, integrability, and plane partitions. *Annales de l’Institut Fourier*, 69(3):1087–1145, 2019. [arXiv:1612.04465](#).
- [71] Michael Wheeler and Paul Zinn-Justin. Littlewood–Richardson coefficients for Grothendieck polynomials from integrability. *J. Reine Angew. Math.*, 757:159–195, 2019. [arXiv:1607.02396](#).
- [70] Michael Wheeler and Paul Zinn-Justin. Hall polynomials, inverse Kostka polynomials and puzzles. *Journal of Combinatorial Theory, Series A*, 159:107–163, 2018. [arXiv:1603.01815](#).
- [69] Peter Finch, Robert Weston, and Paul Zinn-Justin. Theta function solutions of the quantum Knizhnik–Zamolodchikov–Bernard equation for a face model. *J. Phys. A*, 49(6), 2016. [arXiv:1509.04594](#).
- [68] Michael Wheeler and Paul Zinn-Justin. Refined Cauchy/Littlewood identities and six-vertex model partition functions: III. deformed bosons. *Adv. Math.*, 299:543–600, 2016. [arXiv:1508.02236](#).
- [67] Paul Zinn-Justin. Quiver varieties and the quantum Knizhnik–Zamolodchikov equation. *Theoretical and Mathematical Physics*, 185(3):1741–1758, 2015. [arXiv:1502.01093](#).
- [66] Peter Forrester, L. Dang-Zheng, and Paul Zinn-Justin. Equilibrium problems for Raney densities. *Nonlinearity*, 28(7):2265, 2015. [arXiv:1411.4091](#).
- [65] Anita Ponsaing and Paul Zinn-Justin. Type $\hat{C}$ Brauer loop schemes and loop model with boundaries. *Ann. Henri Poincaré*, 3(2):163–255, 2016. [arXiv:1410.0262](#).
- [64] D. Betea, Michael Wheeler, and Paul Zinn-Justin. Refined Cauchy/Littlewood identities and six-vertex model partition functions: II. proofs and new conjectures. *Journal of Algebraic Combinatorics*, pages 1–49, 2015. [arXiv:1405.7035](#).
- [63] Yacine Ikhlef, Robert Weston, Michael Wheeler, and Paul Zinn-Justin. Discrete holomorphicity and quantized affine algebras. *J. Phys. A*, 46(26):265205, 2013. [arXiv:1302.4649](#).

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- [61] Paul Zinn-Justin. Sum rule for the eight-vertex model on its combinatorial line. In *Symmetries, Integrable systems and representations*, volume 40, 2013. Eds Iohara, Morier-Genoud, Rémy. [arXiv:1202.4420](#).
- [60] Roger Behrend, Philippe Di Francesco, and Paul Zinn-Justin. A doubly refined enumeration of alternating sign matrices and descending plane partitions. *J. Combin. Theory Ser. A*, 120(2):409–432, 2013. [arXiv:1202.1520](#).
- [59] R. Rimányi, V. Tarasov, A. Varchenko, and Paul Zinn-Justin. Extended Joseph polynomials, quantized conformal blocks, and a q -Selberg type integral. *J. Geom. Phys.*, 62(11):2188–2207, 2012. [arXiv:1110.2187](#).
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- [57] Alice Guionnet, Vaughn Jones, Dimitri Shlyakhtenko, and Paul Zinn-Justin. Loop models, random matrices and planar algebras. *Comm. Math. Phys.*, 316(1):45–97, 2012. [arXiv:1012.0619](#).
- [56] Paul Zinn-Justin and Jean-Bernard Zuber. Knot theory and matrix integrals. In *The Oxford Handbook of Random Matrix Theory*. 2011. Eds Akemann, Baik and Di Francesco. [arXiv:1006.1812](#).
- [55] Allen Knutson and Paul Zinn-Justin. The Brauer loop scheme and orbital varieties. *J. Geom. Phys.*, 78:80–110, 2014. [arXiv:1001.3335](#).
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- [53] K. Shigechi and Paul Zinn-Justin. Path representation of maximal parabolic Kazhdan–Lusztig polynomials. *J. Pure Appl. Algebra*, 216(11):2533–2548, 2012. [arXiv:1001.1080](#).
- [52] Paul Zinn-Justin. Integrability and combinatorics: selected topics. In *Exact Methods in Low-dimensional Statistical Physics and Quantum Computing*, volume 89. Oxford University Press, 2010. Eds Jacobsen, Ouvry, Pasquier, Serban, and Cugliandolo. <http://www.lpthe.jussieu.fr/~pzinn/semi/intcomb.pdf>.
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- [47] Paul Zinn-Justin. *Six-vertex, loop and tiling models: integrability and combinatorics*. Lambert Academic Publishing, 2009. Habilitation thesis. [arXiv:0901.0665](#).
- [46] Paul Zinn-Justin. Littlewood–Richardson coefficients and integrable tilings. *Electron. J. Combin.*, 16:Research Paper 12, 2009. [arXiv:0809.2392](#).
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